1 Sums and Limits

mathclap & friends

$$X = \sum_{1\le i\le j\le n} X_{ij}$$

$$X = \sum_{\mathclap{1\le i\le j\le n}} X_{ij}$$

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Cramped

$$x^2 \leftrightarrow x^2 \quad {\scriptstyle x^2 \leftrightarrow x^2}$$

Smashoperator

$$V = \sum_{1\le i\le j\le n}^{\infty} V_{ij} \quad X = \sum_{1\le i\le j\le n}^{3456} X_{ij} \quad Y = \sum_{1\le i\le j\le n} Y_{ij} \quad Z = \mathop{T}_{1\le i\le j\le n} Z_{ij}$$

$$V = \sum_{1\le i\le j\le n}^{\infty} V_{ij} \quad X = \smashoperator{\sum_{1\le i\le j\le n}^{3456}} X_{ij} \quad Y = \smashoperator[r]{\sum_{1\le i\le j\le n}} Y_{ij} \quad Z = \smashoperator{\mathop{T}_{1\le i\le j\le n}} Z_{ij}$$

Adjustlimits

$$\text{a) } \lim_{n\to\infty} \max_{p\ge n} \quad \text{b) } \lim_{n\to\infty} \max_{p^2\ge n} \quad \text{c) } \lim_{n\to\infty} \sup_{p^2\ge nK} \quad \text{d) } \limsup_{n\to\infty} \max_{p\ge n}$$

$$\text{a) } \adjustlimits\lim_{n\to\infty} \max_{p\ge n} \quad \text{b) } \adjustlimits\lim_{n\to\infty} \max_{p^2\ge n} \quad \text{c) } \adjustlimits\lim_{n\to\infty} \sup_{p^2\ge nK} \quad \text{d) } \adjustlimits\limsup_{n\to\infty} \max_{p\ge n}$$

2 Tags

$$a = b \tag{Q\&A}$$

See Q&A or is it better with Q&A*? In the star form* `\ref*` *becomes Q&A (*`\refeq*` *is not defined).*

$$a = b \tag*{Q\&A}$$

$$a = b \tag*{[\textbf{Q\&A}]}$$

Normal tags.

$$a = a \tag{1}$$

That was equation (1).

OK tags.

$$a = a \qquad [\mathbf{2}]$$

That was equation [**2**], but recall [**1**]

odd tag.

$$a = a \qquad \{3\}$$

That was equation {3}, but recall {1} and {2}.

weird tag.

$$b = b \qquad ((4))$$

That was equation ((4)), but recall ((1)), ((2)) and ((3)).

Normal tags again.

$$c = c \tag{5}$$

Non-textual

$$d = d \qquad (\mathrm{n}^{th})$$

That was equation (5), but recall (1), (2), (3), (4) and (nth).

$$a = a \tag{6}$$

$$b = b \qquad (**)$$

This should refer to the equation containing $a = a$: (6). Then a switch of tag forms.

$$c = c \tag{7}$$

$$d = d \tag{8}$$

This should refer to the equation containing $d = d$: (8) (but recall (6)).

$$e = e \tag{9}$$

$$f = f \tag{10}$$

$$1 + 1 = 2$$

$$2 + 2 = 4$$

Blabla (2).

3 Arrows

$$A \xLeftarrow[under]{over} B \xRightarrow[under]{over} C$$

$$x \xleftrightarrow[under]{overlooooooong} y \xLeftrightarrow[underlooooooooong]{over} z$$

$$x \xhookleftarrow[bar]{foo} y \xhookrightarrow[bluuuuuuuuuub]{baz} t \xmapsto{heeereeee} k$$

$$k \xleftharpoonup[.]{} l \xleftharpoondown{..} m \xrightharpoondown[...]{} n \xrightharpoonup{....} o$$

$$x \xrightleftharpoons{bluuuuub} y \xleftrightharpoons[blaaaaaab]{} z$$

$$z = \overbrace{\underbracket{x}_{\text{real}} +i \underbracket{y}_{\text{imaginary}}}^{\text{complex number}} \quad \underbrace{1+1}_{=2}$$

4 Matrices

$$\begin{array}{lc} c & cocococococo \\ c & c \end{array}$$

$$\begin{array}{ll} lalalalalala & l \\ l & l \end{array}$$

$$\begin{array}{rr} rererererere & r \\ r & r \end{array}$$

$$\begin{pmatrix} ppppppp & foo \\ l & pppppppppppppppp \end{pmatrix}$$

$$\begin{bmatrix} b & b \\ b & b \end{bmatrix}$$

$$\left\{\begin{array}{lr} B & B \\ B & BBBBBBBrBBBBBB \end{array}\right\}$$

$$\begin{vmatrix} v & v \\ v & v \end{vmatrix}$$

$$\left\|\begin{array}{cr} V & V \\ VVVVVVcVVVVVV & bar \end{array}\right\|$$

$$\left| \begin{smallmatrix} a & blblblbllbblblblblblbl \\ c & d \end{smallmatrix} \right|$$

$$\left[\begin{smallmatrix} a & -b \\ -c & d \end{smallmatrix} \right] \left[\begin{smallmatrix} a & -b \\ -c & d \end{smallmatrix} \right]$$

$$\left\| \begin{smallmatrix} e & -f \\ -g & h \end{smallmatrix} \right\| \left\| \begin{smallmatrix*}[r] e & -f \\ -g & h \end{smallmatrix*} \right\|$$

$$\left[\begin{smallmatrix} a & -bbbbb \\ -c & d \end{smallmatrix} \right] \left[\begin{smallmatrix*}[r] a & -bbbbb \\ -c & d \end{smallmatrix*} \right]$$

$$\left\| \begin{smallmatrix} e & -fffff \\ -g & h \end{smallmatrix} \right\| \left\| \begin{smallmatrix*}[r] e & -fffff \\ -g & h \end{smallmatrix*} \right\|$$

$$\left[\begin{smallmatrix*}[l] a & -bbbbb \\ -c & d \end{smallmatrix*} \right] \left[\begin{smallmatrix*}[r] a & -bbbbb \\ -c & d \end{smallmatrix*} \right]$$

$$\left\| \begin{smallmatrix*}[l] e & -fffff \\ -g & h \end{smallmatrix*} \right\| \left\| \begin{smallmatrix*}[r] e & -fffff \\ -g & h \end{smallmatrix*} \right\|$$

5 Cases

$$\begin{cases} E = mc^2 & \text{Nothing to see here} \\ \int x - 3\,dx & \text{Integral is text style} \end{cases}$$

$$\begin{dcases} E = mc^2 & c \approx 3.00 \times 10^8\,\mathrm{m/s} \\ \int x - 3\,dx & \text{Integral is display style} \end{dcases}$$

$$a = \begin{dcases*} E = mc^2 & Nothing to see here (text in math) \\ \int x - 3\,dx & Integral is display style (text in math) \end{dcases*}$$

$$\begin{rcases} E = mc^2 & 5^6 \quad \text{and so on} \\ \int x - 3\,dx & \int x\,dx \end{rcases} = b$$

$$\begin{rcases} x^2 & \text{for } \int x\,dx > 0 \\ x^3 & \text{else} \end{rcases} \Rightarrow \cdots$$

$$\begin{drcases} E = mc^2 & 5^6 \quad \text{and so on} \\ \int x - 3\,dx & \int x\,dx \end{drcases} = b$$

$$\begin{drcases} x^2 & \text{for } \int x\,dx > 0 \\ \int x^3\,x & \text{else} \end{drcases} \Rightarrow \cdots$$

$$\text{foo} = \begin{cases} \pi & \text{if something} \\ \int \Omega^{\Xi}\,\Omega & \text{otherwise} \end{cases}$$

6 Gathered

$$A = \boxed{\text{first}} B$$
$$\boxed{\text{last}}$$
$$a = b + c$$
$$b = c + d$$
$$\ldots$$
$$\boxed{hello}$$

$$\boxed{f(x) = \int h(x)\,dx}$$
$$= g(x)$$

$$a = b \tag{11}$$

Some text

$$c = d \tag{12}$$

Some short text

$$e = f \tag{13}$$

7 Delimiters

$$|\frac{a}{c}| \quad \left|\frac{a}{c}\right| \quad \Bigg|\frac{a}{b}\Bigg|$$

$$|\frac{a}{b}| \quad \left|\frac{a}{b}\right| \quad |\frac{a}{b}| \quad \Bigg|\frac{a}{b}\Bigg|$$

$$|\pi| \quad |{}_{-}\phi_{-}|$$

$$\left\langle A, \frac{1}{2} \right\rangle \quad \left\langle B \,\middle|\, \sum_k f_k \,\middle|\, C \right\rangle$$

$$\left\{ x \in X \,\middle|\, \frac{\sqrt{x}}{x^2+1} > 1 \right\}$$

$$\langle 1 \mid \tfrac{8}{\frac{4}{1}} \mid 3 \rangle \; \left\langle 1 \,\middle|\, \tfrac{8}{\frac{4}{1}} \,\middle|\, 3 \right\rangle \; \langle 1 \mid \tfrac{8}{\frac{4}{1}} \mid 3 \rangle$$

$$\left(\frac{\pi}{\omega}\right) \cdot \left[\int x dx\right] \ldots [\sqrt{\frac{\sin x}{\cos z}}] \cdots (\frac{\frac{foo}{bar}}{\frac{baz}{qux}})$$

Operators

$$a \coloneqq b \quad a \coloneqq b \quad a \coloneqq b$$

$$a \coloneqq b \quad c \mathrel{::\approx} d \quad e :: f$$

$$\bigtimes \times \nuparrow \ndownarrow \otimes \bigotimes$$

8 Prescripts

$${}^{4}_{12}\mathbf{C}^{5+}_{2} \quad {}^{14}_{2}\mathbf{C}^{5+}_{2} \quad {}^{4}_{12}\mathbf{C}^{5+}_{2} \quad {}^{14}\mathbf{C}^{5+}_{2} \quad {}_{2}\mathbf{C}^{5+}_{2}$$

$${}^{A}_{\mathbf{Z}}\mathrm{X} \rightarrow {}^{A-4}_{\mathbf{Z-2}}\mathrm{Y} + {}^{4}_{\mathbf{2}}\alpha$$

$$a = \frac{\begin{multlined} xy + xy + \int xy \,\mathrm{dx} + xy + xy \\ + xy + xy + xy + xy \end{multlined}}{z} = \frac{\begin{multlined} xy + xy + \int xy \,\mathrm{dx} + xy + xy \\ + xy + xy + xy + xy \end{multlined}}{z}$$

9 Multlines

$$\begin{multline} p(x) = 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3 \\ - 12x^2y^4 - 12xy^5 + 2y^6 - a^3b^3 \end{multline}$$

$$A = \begin{multlined}[t] \boxed{first} \\ \boxed{last} \end{multlined} B$$

$$A = \begin{multlined} \boxed{first} \\ \boxed{last} \end{multlined} B$$

$$A = \begin{multlined}[b] \boxed{first} \\ \boxed{last} \end{multlined} B$$

$$A = \begin{multlined} \boxed{first} \\ \boxed{last} \end{multlined} B$$

$$A = \begin{multlined}[t][12cm] \boxed{first} \\ \boxed{last} \end{multlined} B$$

$$A = \boxed{first}$$

$$\begin{multlined} A = \boxed{first} \qquad\qquad\qquad B \\ \boxed{last} \end{multlined}$$

$$\begin{multlined} \boxed{first} \\ A = \qquad\qquad\qquad \boxed{last} B \end{multlined}$$

$$A = \begin{multlined} \boxed{first} \\ \boxed{last} \end{multlined} B$$

$$\begin{aligned} foo ::= x = 1, \quad x + 1 = 2 \\ y = 2 \end{aligned} \tag{14}$$

$$bar ::= \begin{aligned} x = 1, \quad x + 1 = 2 \\ y = 2 \end{aligned} \tag{15}$$

10 Spread-lines

Spread it

$$\begin{matrix} a & b & c \\ d & e & f \\ g & h & i \end{matrix}$$

$$\begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{pmatrix}$$

$$\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}$$

$$\begin{cases} n/2 & \text{if } n \text{ is even} \\ -(n+1)/2 & \text{if } n \text{ is odd} \end{cases}$$

$$
\begin{aligned}
a &= b + c - d \\
&\quad + e - f \\
&= g + h \\
&= i
\end{aligned} \tag{16}
$$

$$
\begin{multline}
a + b + c + d + e + f \\
+ i + j + k + l + m + n
\end{multline} \tag{17}
$$

$$
a = b \tag{18}
$$

$$
c = d \tag{19}
$$

$$
a_1 = b_1 + c_1 \tag{20}
$$

$$
a_2 = b_2 + c_2 - d_2 + e_2 \tag{21}
$$

$$
\begin{aligned}
a_{11} &= b_{11} & a_{12} &= b_{12} \\
a_{21} &= b_{21} & a_{22} &= b_{22} + c_{22}
\end{aligned}
$$

$$x = y_1 - y_2 + y_3 - y_5 + y_8 - \dots \quad \text{by foo} \tag{22}$$

$$= y' \circ y^* \quad \text{by baz} \tag{23}$$

$$= y(0)y' \quad \text{by Axiom 1.} \tag{24}$$

$$\left.\begin{aligned} B' &= -\partial \times E, \\ E' &= \partial \times B - 4\pi j, \end{aligned}\right\} \qquad \text{Maxwell's equations}$$

$$\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$$

$$\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$$

$$\sum_{\substack{i \in \Lambda \\ 0<j<n}} P(i,j)$$

$$y = ax^2 + bx + c \tag{25}$$

$$f(x) = x^2 + 2xy + y^2 \tag{26}$$

$$\boxed{Firstline}$$

$$\boxed{Secondline}$$

$$L + E + F + T$$

$$R + I + G + H + T$$

$$L + E + F + T$$

$$R + I + G + H + T$$

$$\boxed{WupWup}$$

$$\boxed{Lastline}$$

11 Stepped lines

$$\begin{aligned} 1* \quad x = 1, \quad x + 1 = 2 \quad &\textbf{over} \\ 2* \qquad\qquad\qquad y = 2 \quad &\textbf{over} \end{aligned}$$

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$$\begin{aligned} &\text{See:} s = 2.8, \quad s + 0.2 = 3\text{the end} \\ &\text{See:} t \qquad = \qquad 4.5\text{the} \qquad \text{end} \end{aligned}$$

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12 Shifting equations

$$\begin{aligned} &\boxed{\text{Part 1}} \\ &= \boxed{\text{2nd line}} \\ &19 + \boxed{\text{last part}} \end{aligned}$$

$$\Updownarrow \begin{aligned} \boxed{1} &= \boxed{2} \qquad (27) \\ \boxed{3} &= \boxed{4} \qquad (28) \end{aligned}$$

$$\begin{aligned} a &= b \\ &\vdots \\ &= c \\ &\vdots \\ &= d \end{aligned}$$